1st Place: Aspect 7 – Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

- **Analysis**: This is the most central operational skill for this problem. Students need to directly manipulate a geometric figure (the counter boundary line) drawn on a grid background. They must extract information from the figure (by counting grid squares, measuring) and perform numerical calculations.
- **Manifestation in the Question**: Students need to determine the shape of the counter boundary (an irregular polygon), use the grid as a ruler (each square = 0.5m × 0.5m), obtain the length of each side by **counting the number of grid units along each edge segment** or **decomposing the figure into regular parts to calculate side lengths**, and finally sum these lengths.

2nd Place: Aspect 12 – Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

- **Analysis**: Calculating the total length cannot be done in a single step. It inevitably involves a clear, ordered multi-step process.
- **Manifestation in the Question**: A reasonable problem-solving procedure is:
  1. **Identify and Decompose**: Examine the figure and decompose the outer edge of the counter into several straight line segments (the image is schematic; assume it is primarily a polygon composed of straight segments, possibly with curves).
  2. **Measure/Calculate**: Using the grid, determine the length of each segment (e.g., horizontal/vertical segments by counting squares; diagonal segments may require the Pythagorean theorem or approximation).
  3. **Convert**: Convert the grid unit counts into actual meters (multiply by 0.5).
  4. **Sum**: Add the actual lengths of all segments.
  5. **State**: Provide the total length, likely with units.

3rd Place: Aspect 4 – Make mathematical diagrams, graphs, constructions or computing artifacts and extracting mathematical information from them

- **Analysis**: Although the problem provides the initial diagram, students likely need to actively "create" or manipulate this graphical representation during solving to aid in information extraction.
- **Manifestation in the Question**: Students may need to **mark the diagram** (e.g., label segment lengths, number vertices) or **redraw/decompose the figure on scratch paper** to clearly identify and calculate each side. This process involves systematically extracting the required length information from the given geometric representation.

4th Place: Aspect 8 – Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

- **Analysis**: The problem explicitly requires "Show your work." This means students must clearly present their calculation process, not just the final answer.
- **Manifestation in the Question**: Students should present intermediate steps in their solution, such as: listing the length of each segment (in grid units), showing unit conversion, listing the summation formula. This makes the solution process traceable and verifiable.

5th Place: Aspect 6 – Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms

- **Analysis**: The underlying logic for solving the problem relies on mastery and application of basic geometric concepts.
- **Manifestation in the Question**: Students need to apply the concepts of **length** and **perimeter**. If the edge contains non-horizontal/vertical segments, they may need to implicitly use the **Pythagorean theorem** (right triangle side relationships) or the rule for **distance between two points** to calculate diagonal lengths. This involves employing established mathematical rule systems.

6th Place: Aspect 10 – Make sense of, relate and use a variety of representations when interacting with a problem

- **Analysis**: The problem contains two key representations: a **visual graphical representation** (gridded floor plan) and a **numerical/symbolic representation** (length calculations, final answer). Students need to establish connections between them.
- **Manifestation in the Question**: Students must translate visual information from the diagram (direction of line segments, number of occupied squares) into numerical information (length values) and integrate this numerical information into computational representations.

7th Place: Aspect 13 – Use an understanding of the context to guide or expedite the mathematical solving process, e.g., working to a context-appropriate level of accuracy

- **Analysis**: This is a problem with a real-world context (renovating a shop), and the context itself influences the solution process.
- **Manifestation in the Question**: Students need to understand that "put new edging along the outer edge" implies calculating the **perimeter**, not the area. Furthermore, the context may suggest the answer requires an appropriate level of precision (e.g., to the nearest 0.1 meter or one decimal place), as this is practical for purchasing materials. Understanding the function of "edging" helps identify the correct geometric quantity.

8th Place: Aspect 11 – Switch between different representations in the process of finding solutions

- **Analysis**: This is closely related to Aspect 10 but emphasizes the dynamic process of switching.
- **Manifestation in the Question**: The solution process itself is a series of representation switches: identifying segments from the **visual grid diagram** → recording each side with **numbers (grid counts)** → using **arithmetic operations (multiplication, addition)** for unit conversion and summation → finally producing a **numerical answer with units**. This process may also involve mentally or physically decomposing the figure.

第1位：方面 7 - 处理数字、图形与几何表示

- **分析**：这是本题最核心的操作能力。学生需要直接处理一个几何图形（柜台边界线），该图形绘制在网格背景上。他们必须从图形中提取信息（通过数格子、测量），并进行数值计算。
- **题目体现**：学生需要确定柜台边界的形状（一个不规则多边形），利用网格作为标尺（每格0.5米），通过**数出沿边缘的线段所占的网格单位**或**将图形分解为规则部分计算边长**，来获取每条边的长度，并最终进行求和运算。

第2位：方面 12 - 使用多步骤流程得出数学解答或结论

- **分析**：计算总长度绝非一步完成。它必然涉及一个清晰的、有序的多步过程。
- **题目体现**：一个合理的解题步骤是：
  1. **识别与分解**：观察图形，将柜台外边缘分解为若干条直线段（可能还有弧形？题目中图形为示意图，假设主要为直线段组成的多边形）。
  2. **测量/计算**：利用网格，确定每条线段的长度（例如，水平/垂直线段通过数格子；斜线段可能需要运用勾股定理或近似处理）。
  3. **换算**：将网格单位数转换为实际米数（乘以0.5）。
  4. **求和**：将所有线段的实际长度相加。
  5. **表述**：得出总长度并可能给出单位。

第3位：方面 4 - 制作数学图表、图形或构造物，并从中提取数学信息

- **分析**：虽然题目提供了初始图形，但学生在解题过程中很可能需要主动“制作”或操作这个图形表示，以辅助提取信息。
- **题目体现**：学生可能需要**在图上进行标记**（如标出各段长度、编号顶点），或**在草稿纸上重绘/分解图形**，以便清晰地识别和计算各边。这个过程就是从给定的几何表示中系统提取所需长度信息。

第4位：方面 8 - 阐明解决方案，展示和/或总结、呈现中间数学结果

- **分析**：题目明确要求“Show your work”。这意味着学生必须清晰地展示他们的计算过程，而不仅仅是给出最终答案。
- **题目体现**：学生应在解答中呈现中间步骤，例如：列出各段长度（以格子数计）、展示单位换算、列出求和算式。这使解题思路可追溯、可验证。

第5位：方面 6 - 理解并运用基于定义、规则和形式系统的构造

- **分析**：解决问题的底层逻辑依赖于对几何基本概念的掌握和运用。
- **题目体现**：学生需要运用**长度**和**周长**的概念。如果边缘包含非水平/垂直的线段，可能需要隐含地运用**勾股定理**（直角三角形边长关系）或**两点间距离**的规则来计算斜边长度。这属于运用已建立的数学规则系统。

第6位：方面 10 - 在解决问题时理解、关联和使用多种表征

- **分析**：题目中包含了两种关键表征：**视觉图形表征**（带网格的平面图）和**数字/符号表征**（长度计算、最终答案）。学生需要在两者之间建立联系。
- **题目体现**：学生必须将图形上的视觉信息（线段的走向、占用的格子数）转化为数字信息（长度值），并将这些数字信息整合到计算表征中。

第7位：方面 13 - 利用对情境的理解来指导或加速求解过程

- **分析**：这是一个有现实背景（装修商店）的问题，情境本身会影响求解。
- **题目体现**：学生需要理解“沿柜台外边缘铺设”意味着计算**周长**，而不是面积。此外，情境可能提示答案需要合理的精确度（例如，精确到0.1米或保留一位小数），因为这是购买材料的实际需要。
  对“edging”（镶边）功能的理解帮助确定正确的几何量。

第8位：方面 11 - 在求解过程中在不同表征间转换

- **分析**：这与方面10紧密相关，但更强调动态的转换过程。
- **题目体现**：解题过程本身就是一系列表征转换：从**视觉网格图**中识别线段 → 用**数字（格子数）** 记录每条边 → 用**算术运算（乘法、加法）** 进行单位换算和求和 → 最终得出一个**带有单位的数值答案**。这个过程可能还涉及在头脑或纸面上进行图形分解。